DO NOW, AO1 REVISION

Distance, displacement, speed and velocity

Forces
Paper 2, Combined and Separate
15 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 6 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**Define distance.**	How far an object moves.	☐
2	**Define displacement.**	The straight-line distance from start to finish, with its direction.	☐
3 ★	**Explain the difference between distance and displacement.**	Displacement has a direction (vector); distance does not (scalar).	☐
4 ★	**Explain the difference between speed and velocity.**	Velocity has a direction (vector); speed does not (scalar).	☐
5 ★	**State typical values for walking speed, running speed and cycling speed.**	Walking: 1.5 m/s Running: 3 m/s Cycling: 6 m/s	☐
6	**State a typical value for the speed of sound in air.**	330 m/s	☐
7 ★	**Define velocity.**	The speed of an object in a given direction.	☐
8	**An object moves along a curved path. Describe its speed and its velocity.**	Speed: can be constant. Velocity: changing, because the direction changes.	☐
9 ★	**State what the gradient of a distance–time graph represents.**	Speed	☐
10	**Describe how acceleration appears on a distance–time graph.**	As a curve (the gradient changes).	☐
11	**State the equation linking distance travelled, speed and time.**	distance travelled = speed × time ($s = v\ t$)	☐
12	**Give three factors that affect how fast a person walks, runs or cycles.**	Any three from: age, fitness, terrain, distance travelled	☐

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	QUESTION	ANSWER	✓
13	**State typical speeds for a car on a main road, a train and an aeroplane.**	Car: about 25 m/s Train: about 50 m/s Aeroplane: about 250 m/s	☐
14	**State what a horizontal line on a distance–time graph shows.**	The object is stationary.	☐
15 ★	**Describe how to find the speed at one instant from a curved distance–time graph.**	Draw a tangent to the curve at that time and find its gradient.	☐

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